NEHA DABEERU

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years designing and shipping backend systems at scale — serverless APIs, event-driven pipelines, and distributed workflows for high-traffic, production environments. Track record of root-causing production issues, hardening systems against silent failures, and driving reliability through rigorous testing. AWS Certified Data Engineer.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript
GenAI & ML: AWS Bedrock (Claude/Haiku), Prompt Engineering, RAG, Vector Search, pgvector, FAISS, PyTorch, Transformer Embeddings, Model Serving
Backend and APIs: REST APIs, GraphQL, FastAPI, AWS AppSync, API Gateway, AWS Lambda, VTL Resolvers, Asynchronous Processing, API Design, Authentication, Authorization
Distributed Systems: Event-Driven Architecture, Serverless Architecture, Workflow Orchestration, Message Queues, Idempotency, Retry Strategies, Failure Recovery, State Management, Optimistic Locking, Concurrency Control
AWS and Cloud: Lambda, Step Functions, EventBridge, SQS, S3, DynamoDB, AppSync, API Gateway, OpenSearch, EMR, EC2, IAM, KMS, CloudWatch, X-Ray
Databases and Search: PostgreSQL, MySQL, DynamoDB, Snowflake, Delta Lake, Amazon OpenSearch, HDFS, Hive, Impala
Data Processing: Apache Spark, PySpark, Spark SQL, Databricks, ETL/ELT, Schema Enforcement, Data Reconciliation, Deduplication
Infrastructure and DevOps: Terraform, Docker, GitHub Actions, CI/CD, Git, Infrastructure as Code, Structured Logging, Monitoring, Distributed Tracing, Production Support

PROFESSIONAL EXPERIENCE

Software Engineer, Data & AI Platforms, Merck Sharp & Dohme (New Jersey, USA) Jan 2025 – Present
• Built hybrid BM25 + neural k-NN scoring algorithm with custom normalization, weighted combination, and tie-breaking logic, achieving **99.52%**/**98.46%** auto-coding accuracy on MedDRA/WHODrug terms.
• Root-caused AppSync timeout failures to Databricks cold-start latency (**30s+**); eliminated via async resolver execution, then optimized further with connection caching for reliable sub-timeout responses.
• Implemented immutable audit logging system (DynamoDB Global Tables) providing **100%** traceability of state changes for **2K+** users.
• Designed GraphQL API layer replacing rate-limited REST backend, supporting **20+** concurrent sessions with sub-second response times.
• Developed real-time Mulesoft integration layer connecting internal services to external systems, automating manual data corrections.
• Engineered 12-case E2E test suite (pytest, moto-based AWS mocking) preventing regression of a production data-integrity fix.

Software Engineer II (CONTRACT), Bank of America (North Carolina, USA) June 2024 – Aug 2024
• Built checksum-based reconciliation engine to detect schema drift and silent data mismatches across Cards & Payments DB migration.
• Automated cross-database validation (referential integrity, data-type, record-level checks), cutting manual pre-cutover testing.
• Generated structured exception reports and failure summaries, accelerating migration triage, remediation, & prod cutover decisions.

Software Engineer I, Bank of America (Tamil Nadu, India) June 2020 – Jan 2023
• Orchestrated data processing for financial-risk datasets, parallelizing jobs across sharded workloads to cut pipeline runtime by **30%**.
• Reduced infrastructure costs 30% by redesigning always-on workflows into event-triggered pipelines, eliminating idle resource time.
• Developed reusable data-access components abstracting connection and query logic across database systems, eliminating duplicated integration code for downstream teams.
• Created rule-based translation engine parsing legacy SQL syntax and converting to target dialect, reducing manual effort by 70–80%.
• Strengthened pipeline reliability by making processing steps idempotent and adding checkpointing so failed jobs resume not restart.

SELECTED ENGINEERING PROJECTS

DataFlo & DQ Checker — GenAI Clinical Data Platform: *Python | AWS Bedrock (Claude Haiku) | pgvector | PostgreSQL*
• Built AI-driven field-mapping engine: pgvector similarity search retrieves candidate CDASH matches, Bedrock (Claude Haiku) reranks to confirm best match, reducing study onboarding from 8–12 weeks to 1 week.
• Designed confidence-tiered mapping workflow: high-confidence matches auto-applied, low-confidence matches routed to human review, balancing speed with accuracy for regulatory submission readiness.
• Built GenAI rule engine (DQ Checker) enabling SMEs to author data quality checks in plain language, eliminating manual rule-scripting.

MedPredict — Hospital Readmission Prediction Platform: PyTorch | FAISS | FastAPI | Docker | AWS (EC2/S3)
• Built end-to-end ML pipeline (transformer embeddings, FAISS retrieval, PyTorch classification) trained on **40K+** MIMIC-III records, achieving 92%+ readmission risk accuracy.
• Reduced inference latency to **2s** by deploying real-time prediction API (FastAPI, Docker, AWS EC2/S3) with async handling and model-serving optimizations.

EDUCATION

M.S., Data Analytics, San Jose State University, California Jan 2023 – Dec 2024
B.E., Computer Science and Engineering, SRM University, India June 2016 – May 2020

CERTIFICATION

AWS Certified Data Engineer – Associate, Amazon Web Services May 2025 – May 2028